Your digital twin node, explained

What was built, how the models work, and what the results say

EEE40016 Design of Smart Power Grids
Node: digital twin server (Munir Aslam)
3 October 2026

> This guide explains the whole twin node in plain terms: the question the project asks, how the street and its electricity network are modelled, how a safe charging limit is worked out every five minutes, how bad communications are simulated, the three controllers being compared, and what the final experiments found. Every number comes from the final runs of 2 and 3 October 2026. The project folder is **Downloads\EEE40016\project**.

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How to read this. Section 1 gives the whole story on one page. Sections 2 to 8 explain how the model works, one piece at a time, in the order the pieces are used. Sections 9 to 11 cover the main design fix, the experiments and the results. The rest covers how the model was checked, what went wrong along the way, what the results depend on, what is left, and where everything is. A glossary at the end explains the terms.

1. The short version

The project asks one question: **when does live control of EV charging beat a simple fixed limit, given that the communications between the network and the chargers are never perfect?**

Live control means the network works out a safe charging limit every five minutes from what the street is doing at that moment (a dynamic operating envelope) and sends it to the chargers. A fixed limit is one safe number set in advance and never changed. Live control can give cars more energy when the network has room, but it relies on data and messages that can arrive late, go missing or stop updating. If the data is stale, a limit that looked safe may no longer be safe.

Your node, the digital twin server, answers that question from start to finish. It simulates a real street (55 houses, 22 EVs, rooftop solar and a battery) through an evening from 16:00 to 22:00, degrades the communications in a controlled way, and compares three controllers:

- **A, fixed limit:** one safe limit all evening.
- **B, trust-weighted:** the live limit, turned down as the data becomes less trustworthy. This is the project's proposal.
- **C, binary fallback:** the live limit in full until the data is 15 minutes old, then the fixed limit. This is how deployed systems in the Australian trials work.

What the final experiments found (13 simulated evenings for every setting):

- **Live control pays when the street is congested and the comms are good.** On a normal evening B gave cars up to about 15% more energy than the fixed limit and stayed ahead up to about a minute of delay at low packet loss. Past about two minutes of delay, or with 20% or more packet loss, the fixed limit did as well or better.
- **B is the safer way to do live control.** C got slightly more energy than B on good links, but broke a network limit in 82 runs out of 546, against 9 for B. On the design-day stress case it was 140 against 20.
- **About half of normal evenings are not congested at all,** so live control has nothing to win on them.
- **Frozen data is the most dangerous failure.** Readings that stop changing but keep fresh timestamps fool the trust score completely.
- **A fix made during the work (setpoint schedules)** removed most of the violations caused by late or lost messages, at almost no cost in energy.

> **These are not the project's final results yet.** Three teammates' parts (comms and trust, the charge-point controller, and the battery) are stand-ins written for this build, and the cable ratings are provisional. When the real parts pass the same tests, the experiments are rerun with them.

2. The project and where your node fits

The project has four parts, called nodes, one per team member. Figure 1 shows how they connect. Your node sits in the middle: it holds the model of the street, works out the safe limits and runs the experiments.

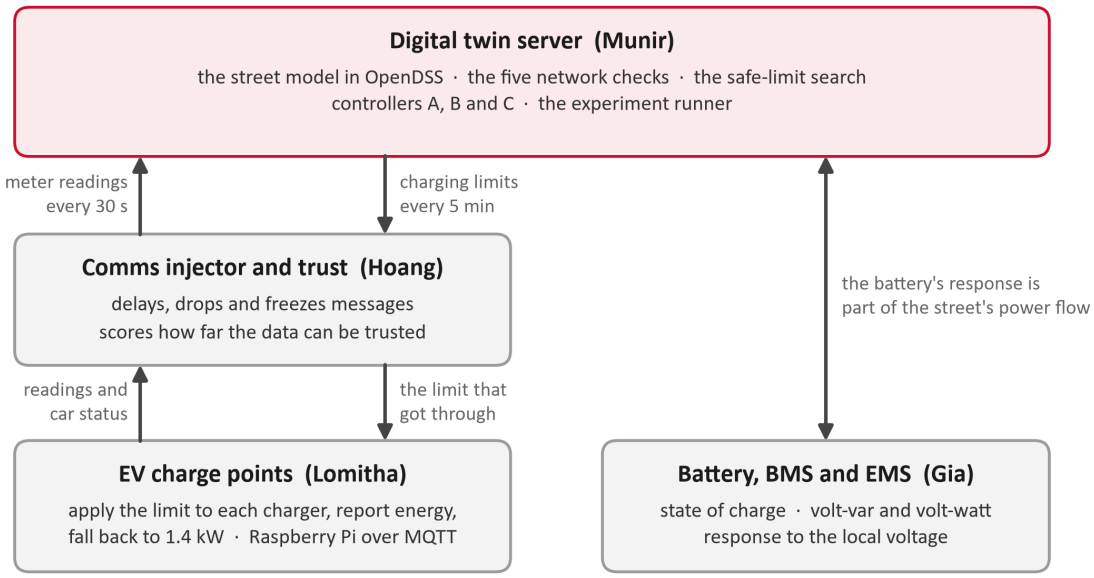


Figure 1. The four nodes. Readings go up through the comms link to the twin; charging limits come back down the same way. The grey boxes are stand-ins written for this build, to be replaced by each teammate's own code.

Node	Owner	What it does	In this build
Digital twin server	Munir	street model, network checks, safe-limit search, controllers, experiment runner	built and validated
Comms injector and trust	Hoang	delays, drops and freezes messages; scores how far the data can be trusted	stand-in, *src/telemetry.py*
EV charge-point controller	Lomitha	applies the limit to each charger, reports energy, falls back to 1.4 kW	stand-in, *src/edge.py*; runs on the Pi
Battery, BMS and EMS	Gia	battery state of charge and inverter response	stand-in, *src/battery.py*

One rule makes the integration work: each teammate writes their logic as a plain Python class with no networking inside it. The simulator calls that class directly for thousands of runs, and the Raspberry Pi runs the very same class behind MQTT for the hardware demo. A test (*test_mqtt_equivalence.py*) runs one evening both ways and checks they give exactly the same answer at every step. It passes at three different comms settings.

The messages between nodes are defined once, in *messages.py*. Since 2 October it is version 2, because a limit now carries a schedule entry and a time window (Section 9). Everyone has to use the new file, and *docs/INTERFACES.md* lists exactly what each person must build.

3. The street: the network model

The street is the **IEEE European Low Voltage Test Feeder**, a published benchmark based on a real suburban street. It has 906 connection points (called buses) and 55 houses fed from one transformer through underground cables. Each house is connected to one of the three phases, which is why the phases can end up unequally loaded.

It is solved in **OpenDSS**, a free power-flow program, as a full three-phase unbalanced model. SINCAL was the first choice, but the laptop's licence only allows 50 nodes, so OpenDSS runs the study and SINCAL is used to cross-check it (Section 12).

Part	How it is modelled
Feeder	IEEE European LV Test Feeder: 906 buses, 55 houses, three-phase unbalanced
Transformer	re-rated from 800 kVA to 150 kVA (4% impedance) so that it can actually be stressed
Supply voltage	1.00 pu at the transformer. The published 1.05 pu left only 0.2 V of headroom below the 253 V limit.
House demand	the measured one-minute load profiles that come with the feeder
Rooftop solar	33 of the 55 houses (60%): 6.6 kW of panels on a 5 kW inverter, clear-sky output for a late-spring day, with a cloud passing from 16:40 to 16:55
Weather	Bureau of Meteorology air temperature (Melbourne Olympic Park), used for the transformer's temperature
Battery	one 100 kWh, 50 kVA unit at the electrical middle of the street (bus 686), responding to local voltage as AS/NZS 4777.2 requires (volt-var and volt-watt)
EVs	22 houses (40%), each with a 7.2 kW charger that can be set in 0.1 kW steps
Time	16:00 to 22:00 in 5-minute control steps (72 steps); meters report every 30 seconds

Is the model right? With the original 800 kVA transformer and no EVs, it reproduces the published IEEE solution at all 2,721 phase-node voltages. The worst difference is 0.0002%, against an allowed 0.5% (validation check V1).

4. One evening, step by step

Every five minutes the same loop runs. Figure 2 shows it, and the numbers below match the numbers in the figure.

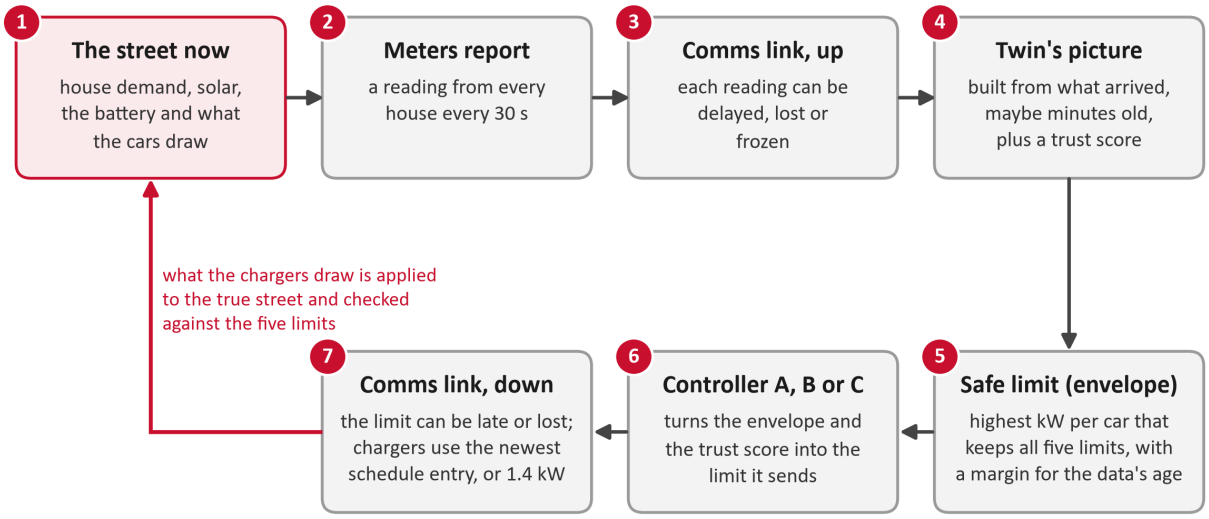


Figure 2. One 5-minute control step. Steps 2 to 6 work from what the twin believes; step 7 is judged against what is really happening.

1. **The street is at its true state:** house demand, solar output, the battery's response and whatever the cars are drawing.
2. **Meters report every 30 seconds.** Each reading carries the value at its own moment.
3. **Each reading goes through the comms link,** which can delay it, drop it or freeze it (Section 6).
4. **The twin builds its picture of the street** from whatever has arrived. That picture may be minutes old. The twin also scores how much it trusts the picture.
5. **The twin works out the safe limit (the envelope)** from that picture: the highest charging power per car that keeps every network limit satisfied, with a safety margin that grows with the age of the data (Section 5).
6. **The controller (A, B or C)** turns the envelope and the trust score into the limit it actually sends (Section 7).
7. **The limit goes back down the comms link,** which can also delay or lose it. The chargers use the newest instructions they hold, or drop to a safe 1.4 kW if those are too old. What the chargers actually draw is then applied to the true street and checked against the five network limits.

The gap between step 5 and the check in step 7 is what the study is about. The limit is worked out from what the twin believes, but it is judged against what is really happening. When the data is stale the two can differ, and a limit that looked safe can break the network. Being more careful costs charging energy. The question is where that trade-off tips.

5. The network limits and the safe limit

5.1 The five limits

Limit	Basis	Value	What it means
Voltage	AS 60038	216.2 to 253 V	every house's voltage must stay inside the legal band
Cable current	cable ratings (provisional, Section 14)	100% of rating	no cable may carry more than its rating; the main cable at the start of the street is rated 175 A per phase
Transformer loading	nameplate	100% of 150 kVA	the transformer must not be overloaded
Transformer hot spot	IEC 60076-7 thermal model	98 °C	the hottest point inside the transformer; it heats up slowly, so a brief overload is tolerated but a sustained one is not
Voltage unbalance	voltage unbalance factor (VUF)	2%	the three phases must not drift too far apart; single-phase chargers push them apart

A run counts only if every limit holds at every one of the 72 steps. Break one limit once and the run is marked unsafe. With realistic demand, the limits that actually bite are **cable current and voltage unbalance**; the transformer rarely limits anything.

5.2 How the safe limit is found

The envelope is one number for each step: the highest power per charger such that, if every plugged-in car drew it, the street would stay inside all five limits. All charging cars get the same limit (equal share). A car that is nearly full only draws what it still needs, and the calculation uses exactly those draws, so the plan matches what the network will really see.

It is found by **bisection**, the same idea as guessing a number with higher-or-lower answers. Try 7.2 kW; if that breaks a limit, try halfway down; keep halving the gap, always on the 0.1 kW steps a charger can actually use. About seven power flows give the exact answer on that grid. A check (V3) compares it with trying every step one by one; the two agree to within 0.1 kW at every test point.

5.3 The safety margin

The envelope is worked out from the twin's picture, which may be old. While readings travel and wait, house demand keeps moving, so the limits used for planning are tightened by a margin that grows with the age of the data. The margins were measured from the feeder's own profiles: each one covers 95% of how far demand drifts over that time.

Age of the data (min)	0	1	2	3	5	7	10	15
Margin on current, voltage, transformer	1.0%	6.8%	8.5%	10.7%	12.7%	14.5%	16.0%	17.1%
Margin on unbalance (percentage points)	0.05	0.33	0.42	0.46	0.53	0.55	0.61	0.75

Unbalance has its own, much larger margin, because one house switching a few kW on one phase moves it a lot. Because the margins cover 95% of the drift, they leave a small tail they do not cover. Most of the remaining violations in the results come from that tail.

5.4 Do no harm

At 16:00 the solar panels are still exporting, and some house voltages can already be above 253 V with no charging at all. Refusing to charge then would be backwards, because charging pulls the voltage down. So when a limit is already broken before any charging, charging is allowed as long as it does not make that breach worse.

6. Communications and trust

6.1 What the comms link does to messages

- **Latency (delay):** each message is delayed by a random amount around the set value. Tested at 0, 5, 30, 60, 120, 300 and 600 seconds.
- **Packet loss:** each message is lost independently with the set probability. Tested at 0, 1, 5, 10, 20 and 40%.
- **Both directions:** readings going up and limits coming down are both degraded, independently of each other.
- **Frozen data:** readings stop changing for 2, 5 or 8 minutes but keep fresh timestamps, as when a meter or gateway hangs. To any check based on age, this looks perfectly healthy.

A late message is not the same as a lost one. The twin only counts a reading as lost once it is older than the delays it actually sees. An early version got this wrong and made the trust score far too low.

6.2 The trust score

The trust score is a number between 0 and 1 for how much the twin believes its picture. It multiplies three simple terms, so it is easy to explain and check:

Term	Formula	What it captures
Age	1 − (age of the newest reading ÷ 15 minutes)	fresh data scores 1; 15-minute-old data scores 0
Loss	1 − measured loss rate	a link that loses messages is less trustworthy
Plausibility	1, or 0.5 if total demand jumped more than five-fold	a crude sanity check on the readings

> **Example.** The newest reading is 3 minutes old and 10% of messages are being lost. Age term 1 − 3/15 = 0.8, loss term 0.9, plausible, so trust = 0.8 × 0.9 = 0.72.

6.3 When instructions stop arriving

The chargers act on the newest instructions they hold. If those are older than 15 minutes, or nothing has arrived, they charge at **1.4 kW**. That fallback matches the 1.5 kW export fallback SA Power Networks uses, and is about the lowest setting a typical AC charger accepts (6 A single-phase). A car that plugged in after the instructions were issued only gets 1.4 kW until new instructions arrive, because those instructions never planned for it. Without that rule, one lost message once pushed a cable to 106%.

7. The three controllers

Controller	Rule	The idea
A, fixed limit	the same limit all evening	the safe baseline; it ignores the data completely
B, trust-weighted	limit = 1.4 + trust × (envelope − 1.4)	lean on the envelope as far as the data deserves, sliding towards the 1.4 kW floor as trust falls
C, binary fallback	full envelope while the data is under 15 minutes old, then the fixed limit	all or nothing, as deployed systems work today

Taking the example from Section 6 (trust 0.72), with an envelope of 5.0 kW and a fixed limit of 2.5 kW:

Controller	Limit sent	Why
A	2.5 kW	always the fixed limit
B	3.9 kW	1.4 + 0.72 × (5.0 − 1.4) = 3.99, rounded down to the 0.1 kW grid
C	5.0 kW	the data is under 15 minutes old, so the full envelope

How the fixed limit A is set. Using 7.2 kW as the fixed limit broke the network on 44 of 72 steps, so that comparison would mean nothing. Instead, for each simulated evening, A is set to the highest fixed value that would have stayed safe all evening, found by trial with perfect information. A real operator could not know that number in advance, so A is if anything a generous baseline.

How results are scored. For B and C, the energy the cars got is compared with what they got under A on the same evening, as a percentage. A run that broke any limit does not count; the results show how many evenings stayed safe.

8. EV demand: who charges, when, and how much

8.1 Why the demand was changed

The first version used placeholder sessions: every car got home around 18:30 wanting 8 to 26 kWh. Checked against real data, that was about three times more charging energy per evening than the evidence supports. With it the transformer was overloaded on 91% of evenings, which is not a realistic street.

8.2 What it is now

When cars get home and how far they drove comes from VISTA, the Victorian travel survey (2024-25): each driver's last arrival home and that day's driving, at 0.17 kWh per km with 90% charger efficiency. That is 1,765 real weekday person-days from Greater Melbourne, weighted to represent the population.

How often people plug in comes from metered trial data. In the Jemena-led Melbourne trial (ARENA, 2023), EVs charged on about a third of days (34%), at about 18 kWh per charging day. So in the **main case** each car plugs in on a given evening with probability 0.34, wanting all the energy driven since it last charged. The **stress case** is the design day Australian distributors plan for (the University of Melbourne method for the ENA): 70% of cars plug in, each wanting two days of driving. You chose these two on 2 October.

Demand	Cars charging (average)	Energy wanted (kWh)	Uncontrolled peak (median)	Evenings over the transformer
Placeholder (old)	22	374	159 kW	91%
Main case	5.9	120	68 kW	0%
Stress case	12.0	171	84 kW	0%

The transformer stays at 150 kVA (about 142 kW). Shrinking it again just to create congestion would not be defensible. With realistic demand, the limits that matter are the cables and voltage unbalance.

Each of the 13 simulated evenings (called seeds) is a fresh random draw: which cars plug in, when, how much they want, and the random delays and losses on the comms link. In the main case, 6 of the 13 evenings never get congested at all.

9. The held-limit problem and the fix

This was the most important fault found during the work, so it gets its own section.

9.1 The problem

The twin sends new instructions every five minutes. If a message is lost or late, the chargers keep using the last one they got. The old code gave each limit a safety margin for the age the data had when the limit was calculated. But a held limit might be used five or ten minutes later, on data that much older, so its margin was too small for how it was actually used. In a replay of the comms link, chargers were running on a held limit on 33% of steps at 120 s of delay even with no packet loss, and on 79% at 300 s.

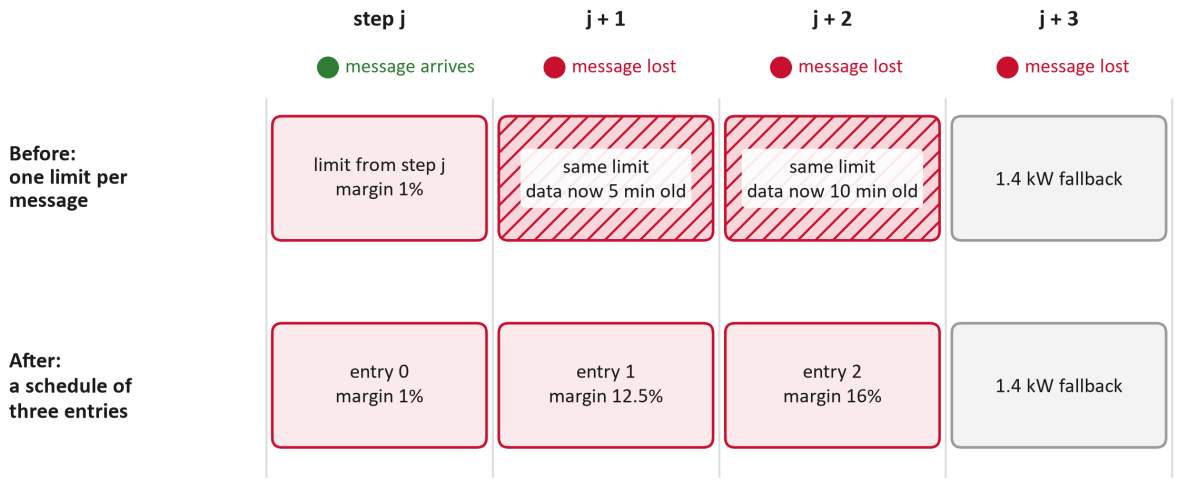


Figure 3. Two lost messages, before and after the fix. Before, the same limit is reused on older and older data; after, each interval has its own entry planned for the data age it will really have.

9.2 The fix: setpoint schedules

The fix was chosen by a design panel that compared three independent designs. Each message now carries a short **schedule of three limits**: one for the next 5 minutes, one for 5 to 10 minutes and one for 10 to 15 minutes. Each later entry is planned with the margin for the older data it will be used on. The chargers use the entry for the current interval, and drop to 1.4 kW when the schedule runs out after 15 minutes, which is exactly when they would have fallen back anyway.

This is how IEEE 2030.5 and CSIP-AUS controls work: each control has a time window, and the device falls back to a default when the window ends. Nothing else changed. The controllers, the trust score and the comms link's random draws are identical, so 10% packet loss means exactly what it meant before.

9.3 How the fix was checked

- 18 runs recorded before the change were reproduced exactly, step by step, with the schedule switched off. With it switched on, the 12 runs that never had a held message were identical too.
- Ten dedicated tests passed, including one showing that entries computed when needed match those a real twin would compute in advance.
- Four independent reviewers checked the code. They found no faults in the control logic; the four bookkeeping issues they did find were fixed.

- The MQTT path still gives exactly the same answer.

9.4 What it achieved

The comparison below uses the same evenings with the schedule off and on. The pass marks were set by the design panel before anything ran. The third column uses the old, heavy placeholder demand, because that is where held limits caused the most trouble.

	Main case	Stress case	Heavy demand
B: violating steps on held limits	39 → 8	67 → 13	139 → 28
C: violating steps on held limits	199 → 86	315 → 117	826 → 226
At least 80% fewer (after setting aside late-arriving and finished cars), B	77%, fail	82%, pass	78%, fail
The same, C	56%, fail	65%, fail	71%, fail
B as safe at 0 s delay as at 5 s	pass	one evening short	pass
B's energy cost on good links under 5%	pass	pass	pass

Violations on held limits fell by about 80% for B and by more than half for C, and B lost less than 5% of its energy on good links. Under the heavy demand, cable overloads fell from 55 to 2 for B and from 638 to 140 for C. The fix did not meet every target the panel set: a held limit is still a little riskier per step than a fresh one, mostly through voltage unbalance on C, which charges right up to the edge with nothing held back. That is reported as it is; the fix was not tuned until it passed.

Two variations were also tried. Making late-arriving cars wait at 0 kW instead of 1.4 kW changed almost nothing. Scoring trust at the moment each entry is used, rather than when it was planned, cost B 15% of its energy to remove just two violating steps. Neither was worth adopting.

10. The experiments that were run

Each run is one simulated evening (72 steps), taking 5 to 30 seconds of computing. The final runs, about 9,000 evenings in all, took roughly eight hours on the laptop in parallel.

Experiment	What changes	Runs
S1, main case	7 delays × 6 loss levels, controllers A, B and C, 13 evenings, main-case demand	1,105
S1, stress case	the same grid on the design-day demand	1,105
S2, frozen data	frozen readings of 0, 2, 5 or 8 minutes at three S1 points (good, moderate and poor links)	351
S3, battery off	the battery removed, on a 3 × 3 part of the S1 grid	130
Fix check	S1 again with the schedule switched off for the main and stress cases, and the old heavy demand run both with and without it	4 × 1,105
Sensitivity	late cars wait at 0 kW; trust scored at the time of use	2 × 156
Timeout	timeouts of 5, 15, 30 and 60 minutes on the 7 congested evenings	672

S4, the warning day (how long the battery keeps critical loads going in an outage), is not built. It needs decisions only you and Gia can make (Section 15).

11. The results

11.1 How to read the result maps

Figures 4 and 5 are grids, one for B and one for C. Each row is a packet-loss level and each column a delay. The number in a cell is the average advantage over the fixed limit, in percent, over the evenings that stayed safe. Blue means ahead of the fixed limit, red means behind, and grey is about even. Where some evenings broke a limit, the cell also says how many stayed safe (for example 8/13 safe). Hatched cells broke a limit in more than half the evenings. The black line is break-even: on the low-delay side of it live control wins, beyond it the fixed limit wins.

11.2 Main case

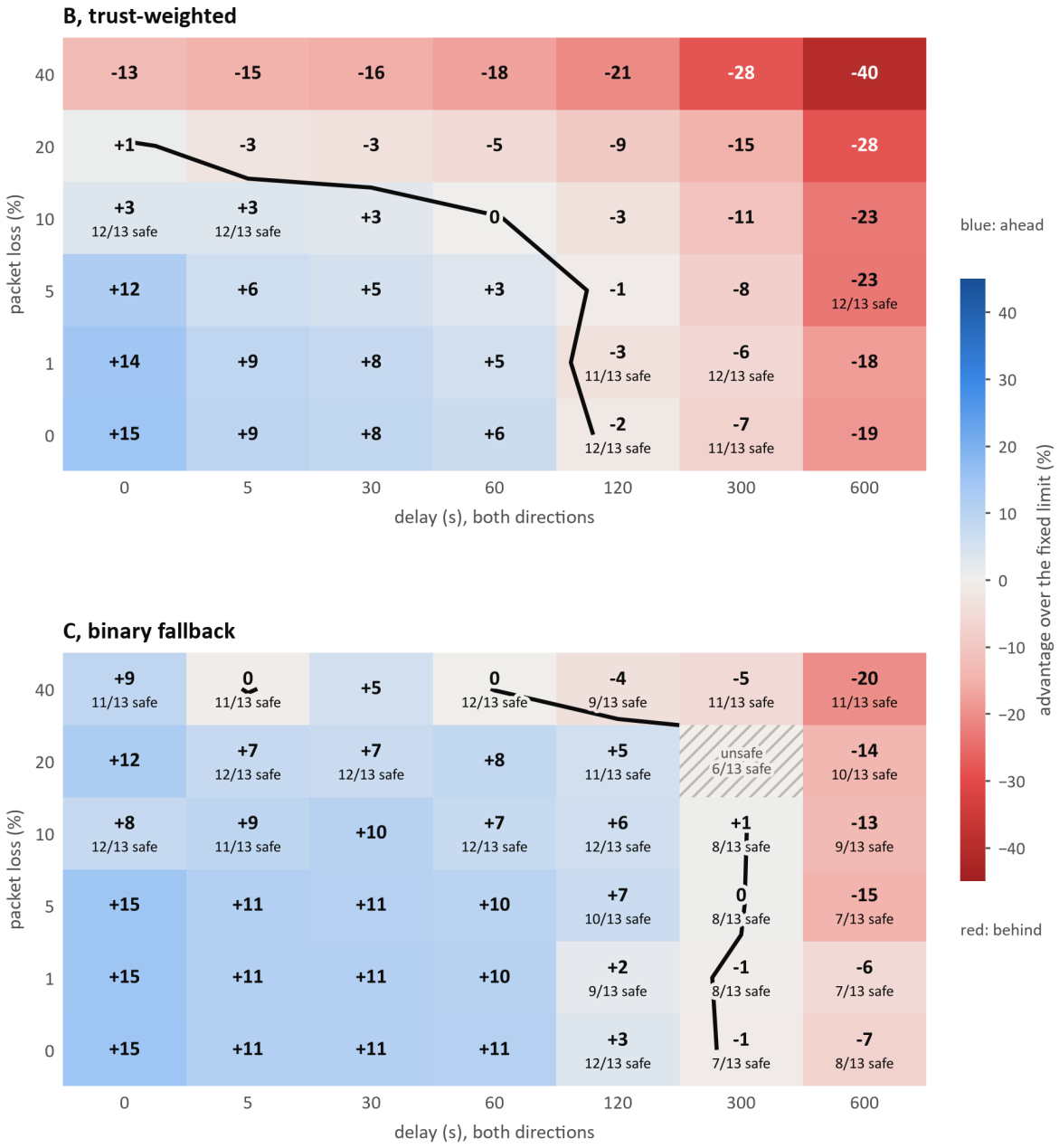


Figure 4. Main case (a third of EVs plug in each evening). Advantage over the fixed limit, averaged over the evenings that stayed safe; 13 evenings per cell. The black line is break-even.

- **B** beat the fixed limit by 15% with no delay or loss, and by 5 to 9% from 5 to 60 s at low loss. It broke even between 60 and 120 s at low loss and at about 5 s once loss reached 20%, and it was behind everywhere at 40% loss. It broke a limit in 9 of 546 runs, all voltage unbalance.
- **C** was 10 to 15% ahead up to 60 s and broke even around 300 s, but it broke a limit in 82 of 546 runs. From 300 s it was unsafe in up to 7 of 13 evenings per cell.
- **A** never broke a limit, by construction. Its limit came out at 2.5 to 6.4 kW on the seven congested evenings and at the full 7.2 kW on the other six, where every controller delivers the same energy. Those six pull every average towards zero.

11.3 Stress case

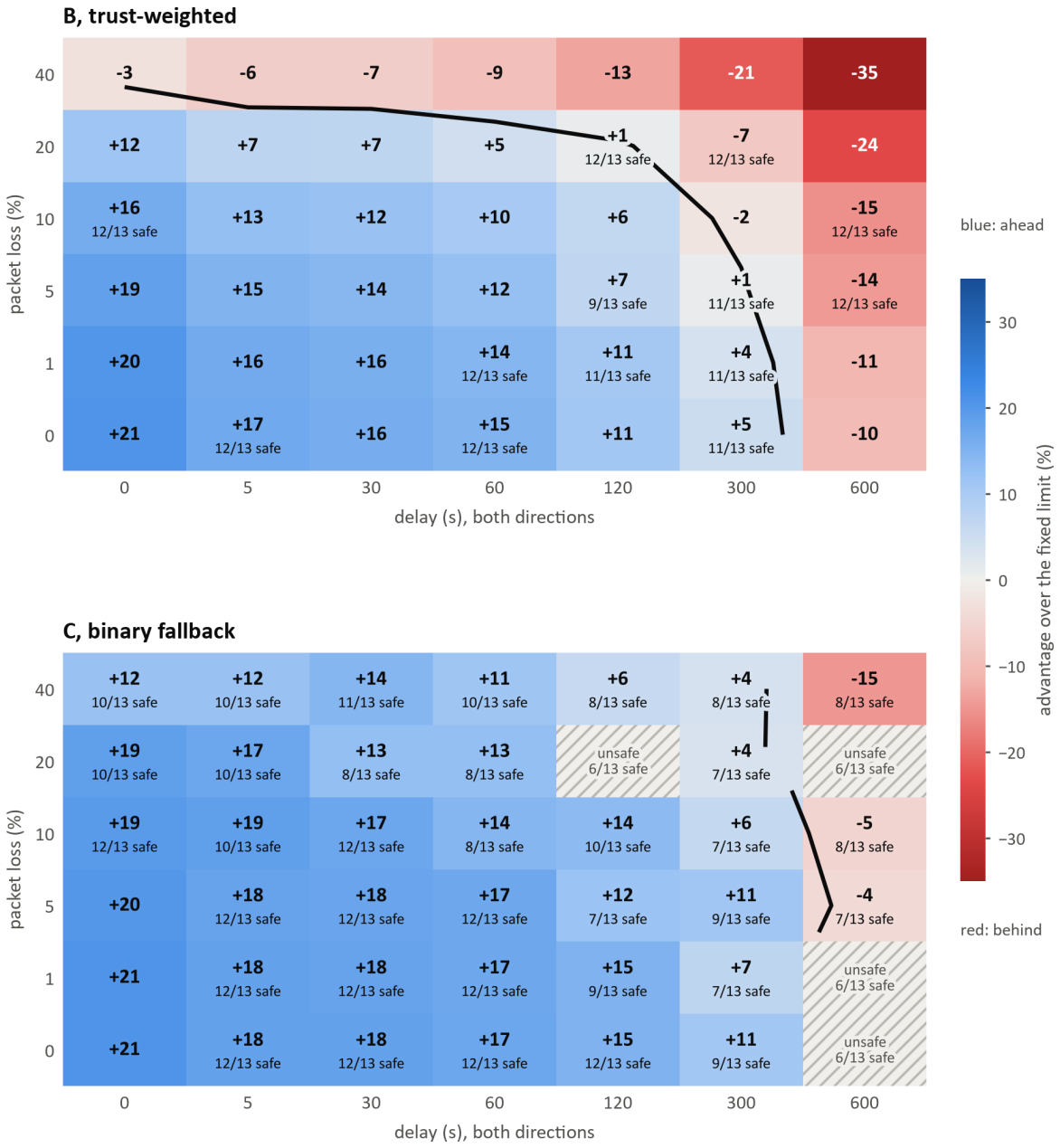


Figure 5. Stress case (the design day, 70% of EVs plug in). Advantage over the fixed limit, averaged over the evenings that stayed safe; 13 evenings per cell.

- Twelve of 13 evenings were congested, most with fixed limits of 2.9 to 4.7 kW, so live control had much more to win.
- B was 21% ahead with no delay, 10 to 17% ahead from 5 to 60 s at up to 10% loss, still ahead at 300 s with 5% loss or less, and behind at 600 s and at 40% loss. It broke a limit in 20 of 546 runs.
- C broke a limit in 140 of 546 runs. Beyond 60 s its cells typically had only 7 or 8 of 13 evenings safe.

The central comparison. On both demand cases, trust-weighting gives up a little energy to binary fallback on good links, and in return breaks network limits seven to nine times less often.

11.4 Frozen data

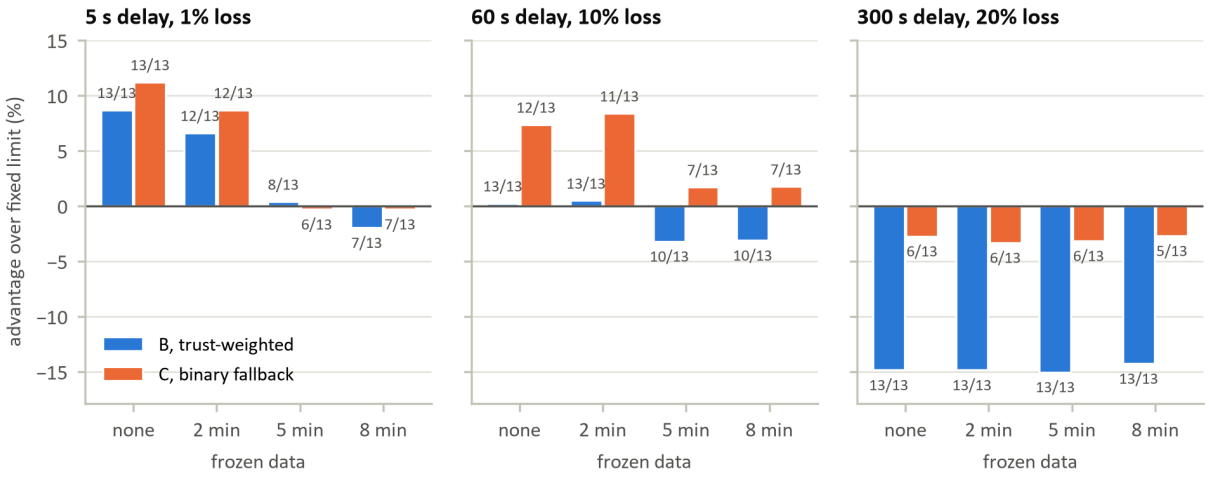


Figure 6. Frozen readings (fresh timestamps, values stuck) at three link qualities, main case. Bar labels: evenings that stayed safe, out of 13.

- A 5- or 8-minute freeze left B safe in only 7 or 8 of 13 evenings on a good link (5 s, 1% loss), and 10 of 13 at 60 s and 10% loss. C fell to 6 or 7 of 13.
- A 2-minute freeze was mostly tolerated (B safe in 12 of 13).
- On a poor link (300 s, 20% loss) the freeze made no difference to B, because its trust was already low.
- The trust score cannot see frozen data that carries fresh timestamps. This bears out the plan's hypothesis that stale data does more damage than packet loss.

11.5 The battery

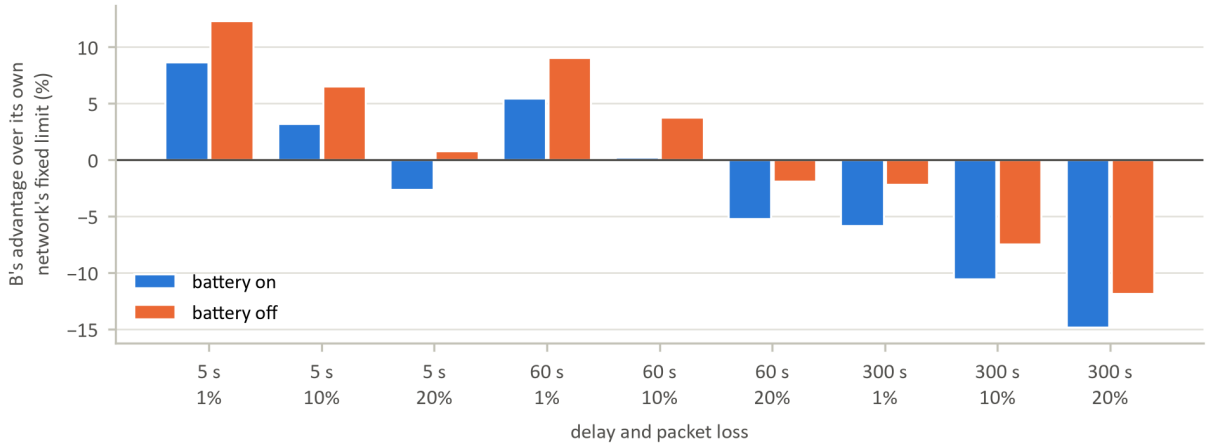


Figure 7. B's advantage over its own network's fixed limit, with the battery (blue) and without it (orange), main case.

Without the battery, the safe fixed limit drops on 3 of 13 evenings (for example from 7.2 to 5.9 kW), and B's advantage over that lower fixed limit grows by 3 to 4 points in every cell. The battery's automatic voltage response mostly helps the fixed limit, so with the battery there is less left for live control to win.

11.6 The timeout

The timeout is how long the system keeps trusting old information before giving up and falling back to 1.4 kW. The study uses 15 minutes; this test tried 5, 30 and 60 minutes as well, changing it at both ends together.

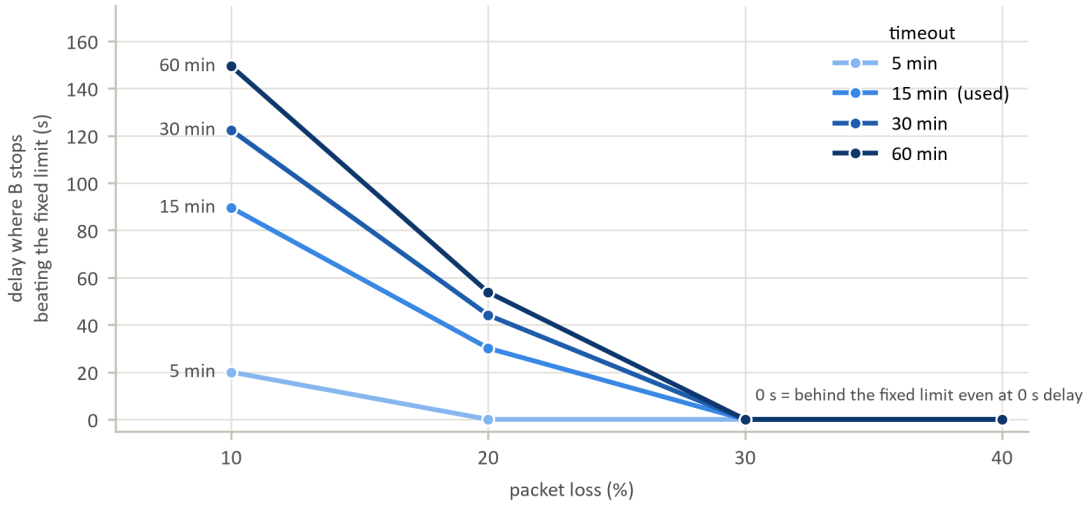


Figure 8. The delay at which B stops beating the fixed limit, for four timeouts (median of the 7 congested main-case evenings).

Packet loss	5 min	15 min (used)	30 min	60 min
10%	20 s	90 s	122 s	149 s
20%	never	30 s	44 s	54 s
30% and 40%	never	never	never	never
Runs that broke a limit	0 of 168	1 of 168	5 of 168	9 of 168

A longer timeout lets B stay ahead at longer delays, but by trusting older data, so it breaks more limits. A very short timeout throws away good information: at 20% loss with a 5-minute timeout, B never beats the fixed limit. The timeout matters less than it did in the earlier model, where one setting kept B ahead past 10 minutes of delay. The report has to state the crossover together with the 15-minute timeout.

11.7 Putting it together

Live control is worth it while the data is fresh: within about a minute of delay and under 10 to 20% packet loss on a normal evening, and within a few minutes on a congested design day. Inside that region, trust-weighting gets most of the benefit of using the envelope in full while breaking limits far less often than the all-or-nothing approach used today. Outside it, a well-chosen fixed limit does as well. The biggest single danger is data that stops updating without anyone noticing.

12. How we know the model is right

The research plan lists nine validation checks (V1 to V9). Each one tests a piece against an answer that does not come from the simulator itself.

Check	What it tests	Result
V1	the network model matches the published feeder	pass: worst 0.0002% (limit 0.5%)
V2	the battery follows AS/NZS 4777.2	pass against the stand-in
V3	the envelope search finds the true envelope	pass: within the 0.1 kW tolerance of a one-by-one search
V4	envelopes are safe and as large as possible	pass: no violations, every envelope maximal
V5	the comms link delivers the delay and loss asked for	pass against the stand-in: within 3.1% and 0.71 points over 20,000 messages
V6	trust behaves sensibly at the extremes	pass against the stand-in: 1 on clean data, 0 after 15 minutes of blackout
V7	the battery conserves energy	pass against the stand-in: balances to 0.0000%
V8	the whole chain gives a known answer	pass: a hand calculation and the simulation agree on the first cable overload, 16:40
V9	results regenerate a week later	pass: all three runs identical eight days after recording (checked 11 October)
MQTT	the hardware path gives the same answer	identical at three comms settings
SINCAL	a second engine agrees	a 47-node cut agrees with OpenDSS within 0.1%; the full street needs the uni computers

V2, V5, V6 and V7 test the stand-ins for now. They double as the hand-over tests: each teammate's real code has to pass the same test before it replaces the stand-in.

V8 in plain words. Every house gets an EV charging at full power with no limit, and simple arithmetic (power divided by voltage on each phase) predicts when the main cable will first overload: 16:40, at 112% of its rating. The full simulation also overloads that cable at 16:40, at 113%. Voltage unbalance breaks earlier, at 16:25, but the hand calculation never tried to predict unbalance, so V8 judges only the cable (amendment 28).

13. Problems found and fixed along the way

All 28 design changes and faults are written up in *notes/plan_amendments.md*, each with its reason. These are the ones that changed the results most.

Problem	What it did	Fix (amendment)
A 7.2 kW fixed limit broke 44 of 72 steps	the research question could not be answered	calibrate A to the best safe fixed limit for each evening (1)
Supply at 1.05 pu left 0.2 V of headroom	the first few cars tripped overvoltage whatever the controller did	1.00 pu, as a distributor would tap down (5)
No safety margin	violations were worst when the data was best	a margin that grows with the data's age (6, 11, 13)
Late messages counted as lost	trust far too low, results biased	count a message lost only once it is older than the observed delays (7)
Envelope planned with cars that were not charging	unbalance violations even with perfect data	plan with exactly the draws the chargers will produce (10)
Unbalance margin a third of its real drift	most fresh-data runs broke unbalance	a separate, larger unbalance margin (21)
A held limit applied to newly arrived cars	one lost message pushed a cable to 106%	a limit covers only the cars connected when it was issued (22)
Random seeds differed between processes	results could not be reproduced	fixed seeding (23)
Held limits planned for fresher data than they had	most live-control violations at long delays	setpoint schedules (24, 25)
SINCAL importer: wrong phase codes, winding type, zero-sequence switch	SINCAL saw 6 kW of a 57 kW street, silently	corrected database codes (26)
Placeholder sessions three times too heavy	the street looked far more congested than it is	VISTA plus metered plug-in rates (27)
V8 compared a cable prediction against any violation	a correct prediction was marked as a failure	V8 judges the limit it predicts (28)

14. What the results depend on

- **Cable ratings are provisional.** The feeder package has no current ratings, so they were estimated from the conductor sizes. They need checking against AS/NZS 3008.1.1. Cable current is one of the two limits that bite, so these numbers matter.
- **Three teammates' parts are stand-ins.**
- **The 15-minute timeout** shapes the crossover (Section 11.6).
- **Margins at the 95th percentile** leave a small tail of risk by design.
- **The demand model.** Each day's driving is drawn afresh, so a heavy driver is not heavy every day, and every charging day is treated as an evening plug-in when the car gets home, which is on the cautious side.
- **Thirteen evenings per setting.** In the main case only seven of them are congested.
- **A steady-state model.** Each 5-minute step is one power flow; anything faster than that is not modelled.

15. What is left, and who does it

Item	Who	Notes
Real comms and trust code	Hoang	pass V5, V6 and the MQTT test; downlink and trust rules R6 and R7 in INTERFACES.md
Real charge-point controller	Lomitha	pass the MQTT test; schedule rules R1 to R5; message version 2
Real battery model	Gia	pass V2 and V7
S4, the warning day	Gia and Munir	decide when the warning arrives, the battery reserve, which loads are critical, when the outage happens and how survival hours are measured
Cable ratings	Munir	check config/cable_ratings.json against AS/NZS 3008.1.1
SINCAL full check	Munir	on the uni computers, using avd_crosscheck.zip
Pi integration	the team	12 October, following Putting_It_Together.pdf; the commands are in team/README.md
Report and slides	the team	agree amendments 1 to 28 with the convener; update the controller labels, the calibrated fixed limit, the demand scenario and the results

Once the real parts pass their tests, the experiments are rerun with them using *tools/run_parallel.py* and *--team all*, and those become the project's final results.

16. Where everything is

Path (inside the project folder)	What is in it
notes/AUDIT.md	every commitment in the research plan, checked against the build, with the final results
notes/plan_amendments.md	the 28 design changes and faults, with reasons
docs/INTERFACES.md	what each teammate has to build
config/study.json	every setting of the study in one place
src/	the model: sweep.py (one evening), envelope.py, network_state.py, controllers.py, telemetry.py, edge.py, battery.py, ev_sessions.py, stages.py
tests/	validation checks V1 to V9, the MQTT test, the schedule tests
tools/	run_parallel.py (the experiment queue), compare_paired.py, the SINCAL tools
results/stages/	main-case results and figures
results/stress_s2/	stress-case results and figure
results/paired_main/, paired_stress/, mechanism_standin/, sensitivity/	the before-and-after checks of the fix
results/timeout_sensitivity/	the timeout test
results/superseded/	old results, kept for the record but not to be quoted
avd_crosscheck.zip	the SINCAL package for the uni computers

17. Glossary

Term	Meaning
Dynamic operating envelope (envelope)	the safe charging limit worked out from the current state of the network and updated regularly, here every 5 minutes
Fixed (static) limit	one safe limit set in advance and never changed
Setpoint	the limit sent to the chargers
Schedule	a setpoint message carrying limits for the next three 5-minute intervals
Fallback	the 1.4 kW the chargers use when their instructions are too old
Latency	how long a message takes to arrive
Packet loss	the share of messages that never arrive
Frozen (stale) data	readings that stop updating while still looking fresh
Trust score	a number from 0 to 1 for how much the twin believes its data
Seed	one simulated evening: a random draw of who charges and what the comms do
Break-even (crossover)	the delay or loss at which live control stops beating the fixed limit

Term	Meaning
Bisection	finding a value by repeatedly halving the range it could be in
Voltage unbalance (VUF)	how unequal the three phase voltages are, in percent
Hot spot	the hottest point inside the transformer's windings
pu (per unit)	a value as a fraction of its nominal value; 1.00 pu is exactly nominal
Bus, phase, feeder	a connection point; one of the three supply conductors; the whole low-voltage street network
OpenDSS, SINCAL	power-flow programs: OpenDSS runs the study, SINCAL cross-checks it
MQTT	the lightweight messaging system the Raspberry Pi uses
IEEE 2030.5, CSIP-AUS	the standard and its Australian profile for sending limits to devices like chargers and inverters
VISTA	the Victorian Integrated Survey of Travel and Activity
Stand-in	working code written in place of a teammate's part until theirs is ready